

Number lines



1

a 2

b -3

c 2

d 9

e -2

f -9

g 4

h -1

i 2

j -6

2 9

3 1

4 -8

5 -2

6 -2

7 3

Integers

8 Their points are the same distance from 0 on the number line, but are on different sides of 0.

9

a i 2

ii -2

b i -12

ii 12

c i 14

ii -14

d i -5

ii 5

10

a -8

b False

11 0

12 16



Comparison and order of integers

13

a 14

b 0

14

a -18

b -1

15

a 13

b -7

c 8

d -40

16

a -21

b -7

c 8

d -40

17 -25, -11, 10, 16, 27

18 15, 11, 3, -9, -13

19

a 4°C

b -17°C

20

a -15°C

b -15°C

21

a $6 > -4$ or $-4 < 6$

b $-4 > -6$ or $-6 < -4$

22

a

i -7 is less than -2

ii True

b

i 7 is greater than -5

ii True

c

i 0 is greater than -1

ii True

d

i 23 is less than -47

ii False

23

a $>$

b $<$

c $>$

d $>$



Applications

24

a Arriving 14 minutes early.

b -14

25

a Cycling 8 km due north.

b -8

26

a i No ii Yes iii No

b -400

27

a 2

b -11

c 10

28

a 14

b -14

29

a -5100

b Being at sea level.

30

a $13 > 8$ or $8 < 13$

b $-10 > -17$ or $-17 < -10$

c $86 > -146$ or $-146 < 86$

d $27 > 17$ or $17 < 27$

e $78.79 > -35.33$ or $-35.33 < 78.79$

f $-71 > -157$ or $-157 < -71$